

DBMS TASK

SQL Query Implementation for E-Commerce Database

Objective: Implement SQL queries for retrieving, filtering, sorting, and reporting e-commerce data.

Database: E-Commerce Database

Main SQL concepts: SELECT • WHERE • ORDER BY • DISTINCT • Filtering • Business Reports

Products • Customers • Categories • Orders • Sales

1. SELECT, WHERE, ORDER BY & DISTINCT

These commands are used to retrieve and organize information from database tables.

```
-- Display all products  
SELECT * FROM Products;
```

```
-- Display selected columns  
SELECT product_name, price  
FROM Products;
```

```
-- Filter products by price  
SELECT product_name, price  
FROM Products  
WHERE price > 1000;
```

```
-- Sort products by price  
SELECT product_name, price  
FROM Products  
ORDER BY price DESC;
```

```
-- Display unique categories  
SELECT DISTINCT category  
FROM Products;
```

2. Product & Customer Search

SQL can search products using price, category, and availability, and retrieve customer details.

```
-- Products in a category
SELECT product_name, price
FROM Products
WHERE category = 'Electronics';

-- Products within a price range
SELECT product_name, price
FROM Products
WHERE price BETWEEN 500 AND 5000;

-- Available products
SELECT product_name, price
FROM Products
WHERE availability = 'Yes';

-- Customer information
SELECT customer_id, customer_name, email
FROM Customers;

-- Find a customer by name
SELECT *
FROM Customers
WHERE customer_name = 'Arun';
```

3. Filtering Conditions

Multiple conditions can be combined using AND, OR, IN, LIKE, and comparison operators.

```
-- Multiple conditions
SELECT product_name, price
FROM Products
WHERE category = 'Electronics'
      AND price < 5000;

-- Search multiple categories
SELECT product_name, category
FROM Products
WHERE category IN ('Electronics', 'Books');

-- Search product names
SELECT product_name, price
FROM Products
WHERE product_name LIKE '%Phone%';

-- Low-stock / unavailable products
SELECT product_name, availability
FROM Products
WHERE availability = 'No'
      OR stock < 10;
```

4. Basic Business Reports & Conclusion

SQL queries can generate simple reports that help an e-commerce business understand products, customers, and sales.

```
-- Average product price
SELECT AVG(price) AS average_price
FROM Products;

-- Highest and lowest price
SELECT MAX(price) AS highest_price,
       MIN(price) AS lowest_price
FROM Products;

-- Number of products by category
SELECT category, COUNT(*) AS product_count
FROM Products
GROUP BY category;

-- Total sales report
SELECT SUM(total_amount) AS total_sales
FROM Orders;

-- Customer order report
SELECT customer_id, COUNT(*) AS total_orders
FROM Orders
GROUP BY customer_id;
```

Conclusion: SQL provides a simple and powerful way to retrieve e-commerce data, apply filtering conditions, search products and customers, and generate useful business reports.

DBMS TASK • END